



PhishGuard AI User Guide

Smart Phishing URL Detection
Powered by AI

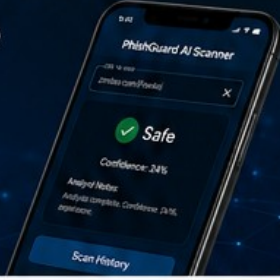


Table of Contents

1. Introduction	1
2. System Requirements	1
3. Running the Backend Server	2
4. Running the Android App	3
5. Using the Application	4
6. Using Share Intent Feature	5
7. Example Results	6
8. Troubleshooting	7
9. Features Overview	8
10. FAQ	9
11. Support	9

Introduction

PhishGuard AI is an Android mobile application designed to detect potentially malicious (phishing) URLs using a machine learning model and a FastAPI backend. You can scan URLs manually or share them from other apps (such as Chrome) using the Android Share Intent feature.

This guide explains how to set up, run and use both the backend and the mobile application.

Key Highlights

- AI-powered phishing detection
- Local scan history storage
- Real-time analysis using FastAPI backend
- Haptic feedback for better user experience
- Share URLs from any app (Share Intent)
- Clean and modern Material 3 UI

2. System Requirements

Backend Requirements

- Windows PC or Laptop
- Python 3.10 or higher
- FastAPI
- Uvicorn
- Internet connection

Mobile Requirements

- Android 10 or newer
- Android Studio (for emulator testing)
- Recommended physical devices for best experience:
 - Google Pixel 3
 - Samsung Galaxy S10
 - Samsung Galaxy S21
 - Google Pixel 7
 - Samsung Galaxy S24

3. Running the Backend Server

Follow these steps to start the FastAPI backend server.

- Step 1** Open Command Prompt or PowerShell.
- Step 2** Navigate to your backend directory:

```
cd C:\Users\lizi\PycharmProjects\PhishGuard_Backend
```
- Step 3** Activate your virtual environment (if any):

```
.\venv\Scripts\activate
```
- Step 4** Run the FastAPI server:

```
uvicorn app:app --host 0.0.0.0 --port 8000
```
- Step 5** Keep this terminal window open while you run the Android app.

Expected Output

You should see output similar to this:

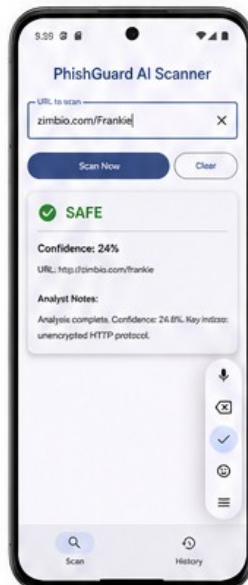
```
Microsoft Windows [Version 10.0.26290.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lizi>cd C:\Users\lizi\PycharmProjects\PhishGuard_Backend
C:\Users\lizi\PycharmProjects\PhishGuard_Backend>.venv\Scripts\activate
(venv) C:\Users\lizi\PycharmProjects\PhishGuard_Backend>uvicorn app:app --host 0.0.0.0 --port 8000
Loading AI model and preprocessing artifacts...
All artifacts successfully loaded. System is operational.
INFO: Started server process [22456]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
```

4. Running the Android App

After the backend server is running, follow these steps to launch the Android application.

- Step 1** **Sync Gradle**
Click the Elephant icon (Sync Project with Gradle Files) at the top right of Android Studio to ensure dependencies are active.
- Step 2** **Start an Emulator or Connect a Device**
You can use an Android emulator or a physical Android device (recommended).
- Step 3** **Run the App**
Click the Green Play button in Android Studio to build, install and launch the app.
- Step 4** **Verify Connection**
Keep the backend terminal open. When you scan a URL, you should see a "POST /predict" log entry in the terminal.



5. Using the Application

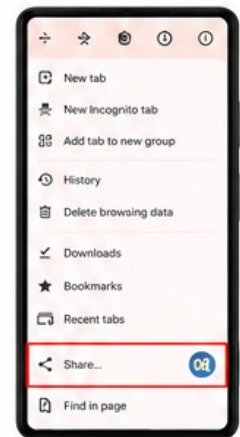
Scan a URL Manually

- Enter a URL in the text field.
Example: <https://google.com>
- Tap **Scan Now**.
- The app will send the URL to the backend and display the result.
- Check the backend terminal – you should see a POST /predict log.

6. Using Share Intent Feature

Share URLs directly from other apps (e.g., Chrome) to PhishGuard AI.

- Open Chrome (or any browser).
- Go to any website.
- Tap the three dots (⋮) in the top right corner.
- Tap **Share...**
- Select **PhishGuard AI** from the list.
- The app will open automatically and start scanning the shared URL.



7. Example Results

URL	True Label	Model Output	Analyst Note (Gemini)
www.softsoft.net/wintopo/	Phishing	Phishing (79%)	Deep subdomain structure and ambiguous TLD. Path pattern consistent with phishing kit deployment.
connecthospital.com	Phishing	Phishing (73%)	Domain impersonates a medical authority. High keyword-to-length ratio typical of social engineering.
secure-login.paypa1.com	Phishing	Phishing (94%)	Typosquatting: digit substitution (1 for l) in a financial brand name. Classic credential-harvest indicator.
google.com/search?q=weather	Benign	Benign (97%)	High-authority domain, standard query parameter structure, no encoding anomalies.
https://mintvelvet.com	Benign	Benign (69%)	Shows ambiguous patterns; exercise caution.
zimbio.com/Frankie	Benign	Benign (24%)	Standard content domain. Short path, no suspicious encoding, registered TLD.
isabellemorin.ndp.ca/contact	Benign	Benign (26%)	Recognised Canadian political domain. Subdomain and path consistent with legitimate site.